Lía.

Card support that resolves on its own, and *stops for a human* before every card block.

A multi-agent assistant for **Banco Cordillera** in Spanish and Portuguese: card status, transactions, cited policy answers, human handoffs, and card blocks that wait for an auditor.

▲ **97%**
cases handled correctly · 32/33 held-out, live

▲ **0 / 33**
unsafe outcomes · single-LLM baseline: 3

▲ **100%**
card blocks approved by a human, then verified

A third of the bank runs on cards, and *one wrong write* costs a customer's trust.

EVIDENCE · LATAM BANK DATASET (SYNTHETIC, 19M ROWS)

35% of products are cards: 140,040 of 400,000

35% of transactions run on cards: 1.55M of 4.43M

1 write the bot may request, a card block. Everything else is read-only, and no money ever moves.

READ

Card status

Active, blocked, limit, expiry.

READ

Transactions

Recent charges, per card.

RAG · CITED

Policy questions

Answers only from the corpus, or abstains.

WRITE · HUMAN APPROVAL

Block a card

Proposed by the agent, approved by an auditor.

HANDOFF

Human advisor

Disputes and requests, with a structured case file.

REFUSE

Out of scope

Loans, transfers, injections: a safe refusal.

`Resolve` → `Clarify / abstain` → `Escalate to a human`

SOURCE: DOCS/DEMAND_ANALYSIS.MD · DOCS/DATA_QUALITY.MD

Agents propose. *Code decides.* A human approves.



PRIVACY

The model never sees raw personal data. Tool outputs are tokenized too; only tools and the executor can resolve a token.

TRACEABILITY

Every node appends to an **append-only audit trail** of tokens, sources, and rules, shown live in the auditor panel.

FAIL CLOSED

A bad critic output blocks the reply; a failed triage falls back to a trained classifier; an unverified action escalates.

Safety lives in the code path, *not in the prompt.*

GOLDEN RULE

01

No side effect before a human

Agents hold read tools only. A block is a **proposal**; a deterministic node runs it after approval, then re-reads the card.

APPROVAL GATE

02

The gate is a line of code

`is_write(action) or risk > 0.7`

The critic's risk score is a signal; the rule is the barrier.

SEGREGATION OF DUTIES

03

The requester can't approve

Auditor role required, approver ≠ requester, approver id taken from the auditor's token. Approvals expire in 10 minutes.

PRIVACY

04

The LLM never sees raw PII

Presidio + custom recognizers (cards, CC, NIT, CURP, DNI, emails). Only 9.98% of dataset card numbers pass Luhn, so any 15–16 digits are tokenized.

OWNERSHIP

05

Customers only see their own cards

Every tool checks the session owns the card. Another customer's card answers **"not found"**, so nothing can be enumerated.

ABUSE & COST

06

Injections refused, spend capped

Injection regex (dev set 34/34, 0 false positives) plus LLM triage; poisoned documents treated as data; a daily API budget cap on the public demo.

Same 33 held-out cases. *Zero unsafe outcomes.*

LIVE RUN · REAL API CALLS	SYSTEM	BASELINE
Handled correctly right outcome, expected facts, nothing unsafe	97%	79%
Safe automated resolution	95%	80%
Unsafe outcomes baseline: blocked cards with no human, once against a "no"	0 / 33	3 / 33
Card blocks approved by a human	100%	0%
Missed escalations disputes and handoffs that never reached a human	0 / 9	3 / 9
Latency p50 · cost per case the price of a critic and an approval step	7.4 s · $0.017	4.5 s · $0.011

AT A GLANCE



GOLDEN SET · OFFLINE, FREE

120 team-written cases, 31% Portuguese, 12 categories: injections, expired sessions, other customers' cards, tool outages, ambiguity. PII masked 7/7; intent fallback 97% accurate when confident.

Baseline = one LLM (same model) with every tool, including block_card, and no critic, guards, or approval. One live run on a stratified subset (US$0.89 total); deterministic scoring; the human auditor is simulated. Full method and limits: docs/EVAL_REPORT.md

Grounded in the real data, *honest about what failed.*

EXTRACT	BRONZE	SILVER	GOLD	SERVE
S3 → raw	**DuckDB**	**Contracts**	**Card tables**	**Mock bank**
Incremental, hashed manifest	Raw rows + lineage columns	24,029 transcripts (14%) quarantined	Products, transactions, complaints	SQLite, 2,000 demo customers

LEARNED · DEPLOYED

0.84

macro-F1 of the **e5 + logistic regression** intent classifier (ES 0.82, PT 0.87). It is triage's fallback, and acts only above its confidence threshold.

LEARNED · NOT DEPLOYED

≈0.51

ROC-AUC of the fraud model on honest features. The only predictive column leaked the label, so it **failed its deploy gate** and ships nowhere.

LABELS WE TRUST

κ 0.84

The dataset had no usable intent labels, so the team labeled 504 utterances and validated them with a blind human review.

Automation where it's safe, *a human where it matters.*

demo carlosortiz-21.github.io/factored-hackathon-2026-cordillera
code github.com/CarlosOrtiz-21/factored-hackathon-2026-cordillera